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A COMPREHENSIVE REVIEW OF BIOPHORUM STANDARDIZED EXTRACTABLES TESTING DATA:

A DEEP-DIVE INTO SIMILARITIES, DIFFERENCES AND TRENDS ACROSS EXTRACTION SOLVENTS AND TIME POINTS

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About BioPhorum

BioPhorum's mission is to create environments where the global biopharmaceutical industry can collaborate and accelerate its rate of progress, for the benefit of all.

Since its inception in 2004, BioPhorum has become the open and trusted environment where senior leaders of the biopharmaceutical industry come together to openly share and discuss the emerging trends and challenges facing their industry.

Growing from an end-user group in 2008, BioPhorum now comprises over 90 manufacturers and suppliers deploying their top 3,500 leaders and subject matter experts to work in seven focused Phorums, articulating the industry's technology roadmap, defining the supply partner practices of the future, and developing and adopting best practices in drug substance, fill finish, process development and manufacturing IT. In each of these Phorums, BioPhorum facilitators bring leaders together to create future visions, mobilize teams of experts on the opportunities, create partnerships that enable change and provide the quickest route to implementation, so that the industry shares, learns and builds the best solutions together.

Abstract

Extractable and leachable (E&L) compounds associated with the use of polymeric single-use systems (SUS) are of primary concern due to their direct impact on product and patient safety. The 2014 Biophorum extractables protocol provided a standardized framework to conduct E&L testing and has been widely used by SUS suppliers as well as the end-user biomanufacturing companies. Since its implementation, several concerns associated with the testing methods proposed in the protocol have been raised by testing laboratories. A comprehensive review of the data generated using the 2014 BioPhorum extractables protocol was initiated by Biophorum's E&L workstream to mitigate these concerns.

This document aims to summarize the findings of the data review and make scientific, data-driven recommendations to address specific concerns about extraction solvents and testing timepoints. This review established that certain extraction solvents (5M Sodium Chloride and 1% Polysorbate 80) and timepoints (Time 0) proposed in the testing matrix of the 2014 protocol added limited value to determination of the full extractables profile. Such trends were observed for organic and elemental extractables alike. Based on these findings, specific recommendations about elimination of these solvents and timepoints are outlined in this document.

1.0

Introduction

The proliferation of single-use systems (SUS) in biopharmaceutical manufacturing operations has been due to a combination of factors, including the increased demands on throughput efficiencies and flexibilities stemming from a growing number of therapeutic modalities manufactured at biopharmaceutical facilities. Due to their polymeric materials of construction, the requirements for extractables and leachables (E&L) testing of SUS were established by regulatory agencies[1,2]. These requirements formed the basic framework of the harmonized extractables protocol developed by the BioPhorum E&L Workstream in 2014[3].

The protocol, as originally developed, proposed an all-encompassing testing matrix consisting of various extraction solvents and testing time points under accelerated temperature conditions to represent majority of the actual processing conditions currently used in the industry[4]. Since it was published, the 2014 BioPhorum E&L protocol has been implemented by several SUS supplier and end-user organizations. A wide variety of components ranging from upstream to final drug product consumables have been tested using the 2014 BioPhorum extractables protocol.

As the 2014 BioPhorum extractables protocol gained credence with regulatory agencies and the biomanufacturing industry worldwide, questions and concerns about the implementation of the proposed testing matrix were raised by several test laboratories and biomanufacturers. Some notable concerns included analytical interference for specific extraction solvents, instrument limitations for elemental extractables testing and the resources associated with implementing the full BioPhorum testing matrix.

Issues such as analytical interference and instrument limitations can lead to confounding data, thereby making it difficult for biopharmaceutical manufacturers to evaluate the potential risk of extractables toxicity. Furthermore, the resources required to perform extractables testing for numerous SUS components across a broad group of biopharmaceutical companies could lead to testing redundancy within the industry and longer lead times, putting the consumable supply chain at greater risk.

To address these valid concerns, the BioPhorum E&L workstream evaluated currently available extractables data collected using the 2014 BioPhorum protocol to assess the suitability of various extraction solvents and time points. This document summarizes the findings of the comprehensive extractables data review, lists any specific similarities or differences observed across various solvents and time points, and determines the associated risks in terms of SUS qualification and patient safety. The findings outlined herein form the basis of the updated Biophorum extractables testing protocol[5].

2.0

Background

The 2014 BioPhorum protocol was originally developed to fill the gap between the general regulatory expectations for extractables testing and biomanufacturers' varying testing practices performed to support different SUS applications. A survey from 17 major BioPhorum members companies across 26 different sites covering a broad range of SUS components and their applications[4], combined with BioPhorum E&L workstream's discussions with health authority representatives, was used to propose the standardized methods stated in the BioPhorum protocol.

The 2014 extractables protocol testing matrix for various SUS components is shown in Table 1.

Table 1: Current extractables protocol test conditions for single-use systems

	Solvents						Time				
	50% Ethanol	1% PS-80	5M NaCl	0.5N NaOH	0.1M Phosphoric acid	WFI[a]	Time 0 (≤30 min)	24 hours	7 days	21 days	70 days
							Temperature				
							Ambient (25°C)	40°C			
Storage, mixing, and bioreactor bags	X	X	X	X	X	X	X	X		X	X[b]
Tubing	X	X	X	X	X	X	X	X		X	X[b, c]
Tubing connectors and disconnectors	X	X	X	X	X	X	X	X		X	
Aseptic connectors and disconnectors	X	X	X	X	X	X	X	X	X		
Sterilizing-grade filters/process filters	X	X	X	X	X	X	X	X	X		
Tangential-flow filtration cassettes	X	X	X	X	X	X	X	X		X	
Sensors and valves	X	X	X	X	X	X	X	X		X[d]	
Chromatography columns; elastomeric parts (gaskets, O-rings, diaphragms and septum); wetted polymeric surfaces of positive displacement pumps	X	X	X	X	X	X	X	X			

Abbreviations: PS-80 – Polysorbate 80; WFI – water for injection; min – minute

[a]Deionized water can be used for this purpose if WFI is not available.

[b]Duration, specified for testing storage bags and tubing, is necessary to support 3-year storage time at 0 °C.

[c]Tubing is included because tubing sections are typically integrated with bags during storage.

[d]The 21-day time-point applies only to sensors used with bioreactors (e.g. for dissolved oxygen and pH).

Extraction solvents, exposure temperature and time points were selected to represent the worst-case conditions of typical biomanufacturing processes. The Time 0 (≤30 minutes) time point was included to cover any potential initial volatile extractables and the Time 70-day time point was specified to support the long-term storage of the drug substance, drug product, buffers and other process stream solutions. Similarly, the water for injection (WFI), 0.1M phosphoric acid (H_3PO_4, acid, low pH) and 0.5N sodium hydroxide (NaOH, base, high pH) solvents were included to represent the typical buffer-based manufacturing processes. The model solvent chosen to represent the high ionic strength solutions used in bioprocessing was a 5M sodium chloride solution (5M NaCl). The 50% ethanol (EtOH:H2O) solvent was selected to represent the typically used organic solvents in biomanufacturing processes and 1% Polysorbate 80 (PS-80) solvent represented surfactant-containing process solutions.

3.0

Scope

Based on the feedback from various testing laboratories, SUS suppliers and biopharmaceutical manufacturing companies, the data review process primarily focused on answering the following questions:

- Are all six extraction solvents value-added?
- Are all testing time points value-added?
- Is there an opportunity to reduce the analytical challenges associated with extractables studies based on scientific justification?

Although testing costs were listed as one of the potential concerns, a cost-benefit analysis was not included in the scope of this review. All data sets were evaluated for scientific data analysis purposes only.

4.0

Data sharing

Data sharing was an integral component of the data review process. To provide the opportunity for an unbiased data review and trend analysis, consumable suppliers were asked to share data generated per BioPhorum's 2014 extractables protocol. A team of members from five different end-user organizations was formed to review the shared data.

However, due to inherent data ownership and privacy issues associated with the business processes involving data sharing, a six-way confidentiality agreement would be required for each supplier sharing the data. As such arrangements often tend to be time-consuming, an alternate anonymized blinded approach (referred to as the 'simplified data format' in this document) was developed for collaboratively sharing the data with the review team. With this approach, all confidential aspects (such as supplier name, materials of construction, component catalog information, peak identities, etc.) were blinded before providing the data to the review team. The suppliers were identified using an alphabetical naming convention (e.g. Supplier A, B, C, etc.) and the test components were identified using a broad classification of component types (e.g. filter, bag, tubings, etc.). No specific information regarding testing laboratories, components attributes or test method details were provided.

5.0

Review methodology

The review focused on the representative extractables data packages from the various polymeric single-use components shown in Table 2 to evaluate the extraction solvents and study time points specified in the 2014 BioPhorum protocol. In particular, the data review focused on the following questions:

For organic extractables

- 'Unique peaks' vs. 'overlapping peaks' from various solvents
  - Are the observed extractable peaks unique to one particular solvent?
  - Are these observations true across all material components?
- Time point trends observed at 0 day (<30 minutes) and 70 days
  - Are the Time 0 and time 70-day timepoints exhibiting unique extractable peaks?
  - Is the continued testing of 0-day and 70-day time points value-added?

For inorganic extractables (elemental extractables)

- Extraction solvents
  - Which solvent(s) yield the highest level and/or highest number of elemental extractables?
  - Which solvent(s) exhibited the highest number of ICH Q3D-relevant elemental extractables?

Table 2: Extractables data packages (n=28) summary provided by the suppliers included in the scope of the data review[i]

Supplier	Component type					
	Film bags	Tubing/tubing kit	Small components			Filter
			Aseptic connector/disconnector (gamma-irradiated)	HDPE port	Sterile connector (autoclaved)	
A	3					
B	5	3	2			
C	1					
D	1		2	1	1	2
E	1		1			1
F	1					2
G						1
Total	**12**	**3**	**7**			**6**

[i]Data sets were provided by two additional suppliers not listed in Table 2. These data sets were not included for this review as the test components were considered out of scope of a typical SUS for biomanufacturing process and the data were not available in the 'simplified data format'

In general, each data set from the 28 data packages was provided in the simplified data format, as shown in Figure 1. Each data set contained all data from the various analytical methods to represent VOC (volatile organic compounds), SVOC (semi-volatile organic compounds), NVOC (non-volatile organic compounds), and elemental extractables. Each identified extractable peak within a given dataset was adjusted for the respective blank/control samples (i.e. a solvent background was subtracted from the sample peak to represent the true extractable peak contributed by the single-use component). Each observed extractable peak was anonymized and labeled numerically as 'peak 1, peak 2, etc.'. This allowed a visualization of trends for the number and concentration of either unique or overlapping peaks, observed across various solvents and time points for a given component.

The number of unique and overlapping extractable peaks were identified for high-level analysis, as shown in Figure 1. This allowed an unbiased, holistic evaluation of data by not focusing on a particular supplier or component type.

Figure 1: Simplified data format and analysis approach for organic extractables showing examples of a unique peak and an overlapping peak

Organic Extractables

Peak	Time Point	WFI	NaCl	H_3PO_4	NaOH	PS-80	EtOH:H_2O
1	T0	-	-	-	-	-	-
	24 Hr	-	-	-	-	-	1.3
	21 Days	-	-	-	-	-	2.3
	70 Days	-	-	-	-	-	3.3
2	T0	-	-	-	-	-	-
	24 Hr	-	-	-	0.06	-	0.27
	21 Days	-	-	-	0.09	-	0.43
	70 Days	-	-	-	0.46	-	0.44

Peak 1 unique to EtOH:H_2O

Peak 2 overlaps between NaOH and EtOH:H_2O

Redacted peak identity as simple numbers

Reported concentration (already accounted for control/blank)

Original content required

6.0

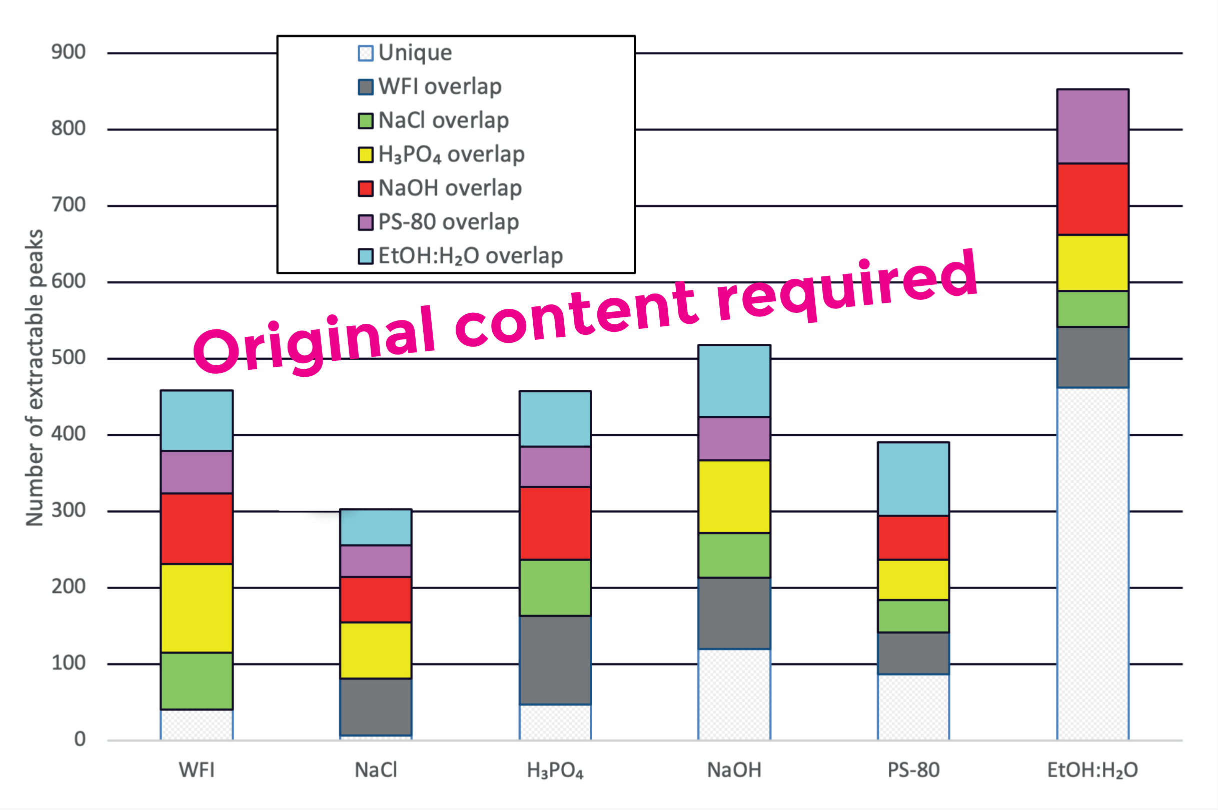
Data evaluation

As shown in Figure 1, during data evaluation, an extractable peak observed in only one extraction solvent was denoted as a unique extractable, and an extractable peak observed in multiple (two or more) extraction solvents was denoted as an overlapping extractable. To evaluate the value of all six solvents, the focus was on unique extractable compounds since these would indicate if a given solvent would produce extractables that were not covered by other solvents.

6.1 Organic extractables

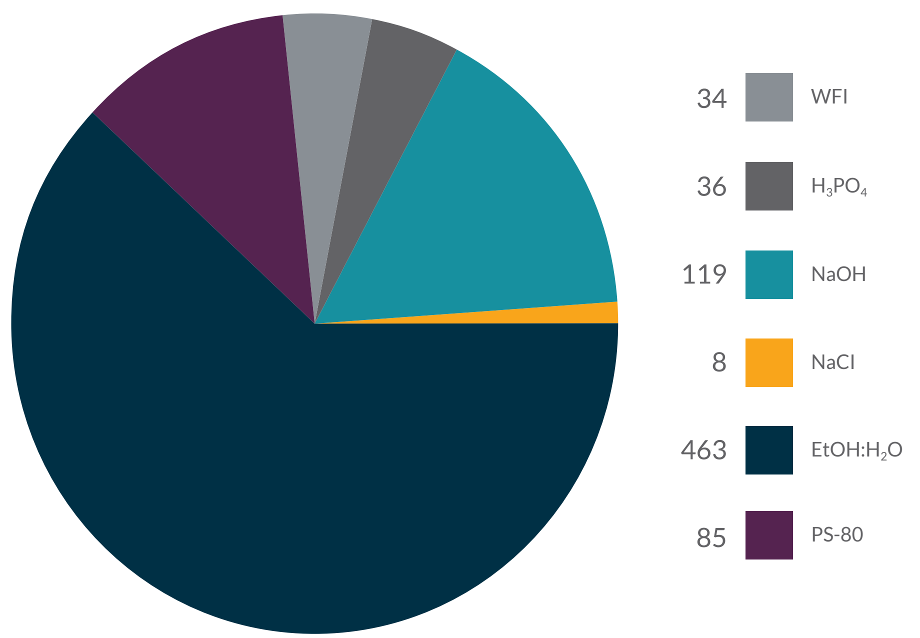
Only one dataset per component was included for review of trend analysis across the six solvents. If additional data on a second lot of the same component and duplicate data on a single lot of components were provided by the supplier, the dataset presenting the highest number extractable peaks (interchangeably referred to as 'extractables' in later sections) was used to represent the worst-case scenario. A total of 1,168 extractable peaks were observed in the 28 data packages received from suppliers. An overall distribution chart of unique vs. overlapping peaks across the six extraction solvents is shown in Figure 2.

Figure 2: Summary of unique and overlapping extractable peaks across the six extraction solvents



Of the total observed extractable peaks, 745 extractables (corresponding to 64% of the total observed extractables) were found to be unique, i.e. observed in a single extraction solvent. The distribution of the 745 unique extractables per solvent is shown in Figure 3. H_3PO_4 overlap.

Figure 3: Distribution of unique extractables by extraction solvent



Of the 745 unique extractables, the majority were observed in the organic extraction solvent (50% $EtOH:H_2O$ mixture). A moderate number of extractables were observed in WFI (neutral), 0.1M H_3PO_4 (acidic solvent) and 0.5N NaOH (basic solvent), with 34, 36 and 119 extractables in each of these solvents, respectively. These solvents amounted to approximately 25% of unique extractables and approximately 16% of the total observed extractables across the pH range. As these four extraction solvents represented approximately 87% of all unique extractables, it was determined that these solvents provided valuable information and were an essential part of the standardized testing matrix.

Only eight unique extractables were observed in the 5M NaCl solvent, corresponding to 1% of the unique extractables and less than 1% of the total observed extractables. Based on this analysis, it was determined that the high ionic strength solvent (5M NaCl) added limited value to the standardized extractables testing matrix. Therefore, eliminating the NaCl extraction solvent from the testing matrix would present minimal risk of missing material-specific extractables.

A total of 85 unique extractables were observed in the 1% PS-80 solvent corresponding to 11% of the unique extractables and 7% of the total observed extractables. According to this analysis, PS-80 appeared to be valuable as an extraction solvent. However, based on feedback from testing laboratories, suppliers and end-users, the complexities of testing with PS-80 were further reviewed. It is generally acknowledged that PS-80 is a challenging extraction solvent since it is known to readily degrade, primarily by oxidation and hydrolysis. Differentiating PS-80-related extractables from non-extractables or degradation products has been difficult since PS-80 has historically been observed to degrade differently within test and control samples, between time points, between extraction solvent preparations using PS-80 from different lots and even with the use of different PS-80 bottles from the same lot[6,7]. This could potentially lead to PS-80 degradation products being mislabeled as extractables (false positives).

To evaluate the number of potential false positives in the PS-80 data sets, each supplier was asked if they had excluded the PS-80 degradation products (false positives) from their unique PS-80 extractables. Of all suppliers that had provided data sets, only 1/3rd of the suppliers confirmed that PS-80 degradation peaks (false positives) were identified and eliminated from their respective data sets. The remaining six suppliers acknowledged that PS-80 degradation products were not eliminated from the data sets.

The total number of unique extractables from the six suppliers who had not removed PS-80 degradation was 70. Assuming that all 70 compounds were PS-80 degradation products and not true extractables, the total number of unique extractables for the PS-80 extraction solvent was reduced from 85 to 15, corresponding to 2% of the unique extractables and 1% of the total observed extractables. Conversely, if all PS-80 compounds from all suppliers were assumed to be true extractables and not false positives, the total number of PS-80 unique extractables amounted to 85, corresponding to 11% of the unique extractables and 7% of the total observed extractables. Based on this assumption range, the actual number of unique extractables from the data sets would be between 2–11%.

To further understand the differences between PS-80 false positives and unique extractables, individual interviews were conducted with each of the suppliers to determine the identity of the observed PS-80 extractables. In particular, the identification of peaks was obtained for data sets from the suppliers that did not eliminate the PS-80 degradation peaks. It was noted from the peak identities that the majority of the peaks originally labeled as unique PS-80 extractables from suppliers were indeed known PS-80 degradation products and not true extractables from the individual materials. Not accounting for the known degradation peaks from all data sets, it is expected that the total number of unique PS-80 extractables would be closer to the lower end of the 2–11% assumption range.

Furthermore, it was observed from the peak identities of the overlapping extractables that the majority of the extractables were non-polar compounds such as long-chain alkanes, fatty acids and esters, antioxidants and their degradation products[ii]. These were observed in slightly higher amounts in the PS-80 solvent compared to other solvents. However, such compounds are generally known to be less toxic and have higher permissible safety tolerances.

After a thorough review of the overall PS-80 extractables data and degradation peaks, it was concluded that the total number of unique extractables is relatively low. Also, the likelihood of observing any highly toxic extractables in this solvent was minimal, as demonstrated from the PS-80 extraction datasets across a broad range of common materials of construction in the six categories of SUS components shown in Table 2. Therefore, eliminating this solvent from the standardized testing matrix presents minimal risk to product safety and of missing material-specific extractables.

[ii]The general chemical class was assessed only for those extractable compounds that were identified by the suppliers at the time of the associated extractables study. Individual peak identities are not disclosed in this paper for the purposes of maintaining confidentiality. End-users may obtain such information directly from the suppliers on a case-by-case basis.

6.2 Elemental extractables

6.2.1 Data processing guidelines

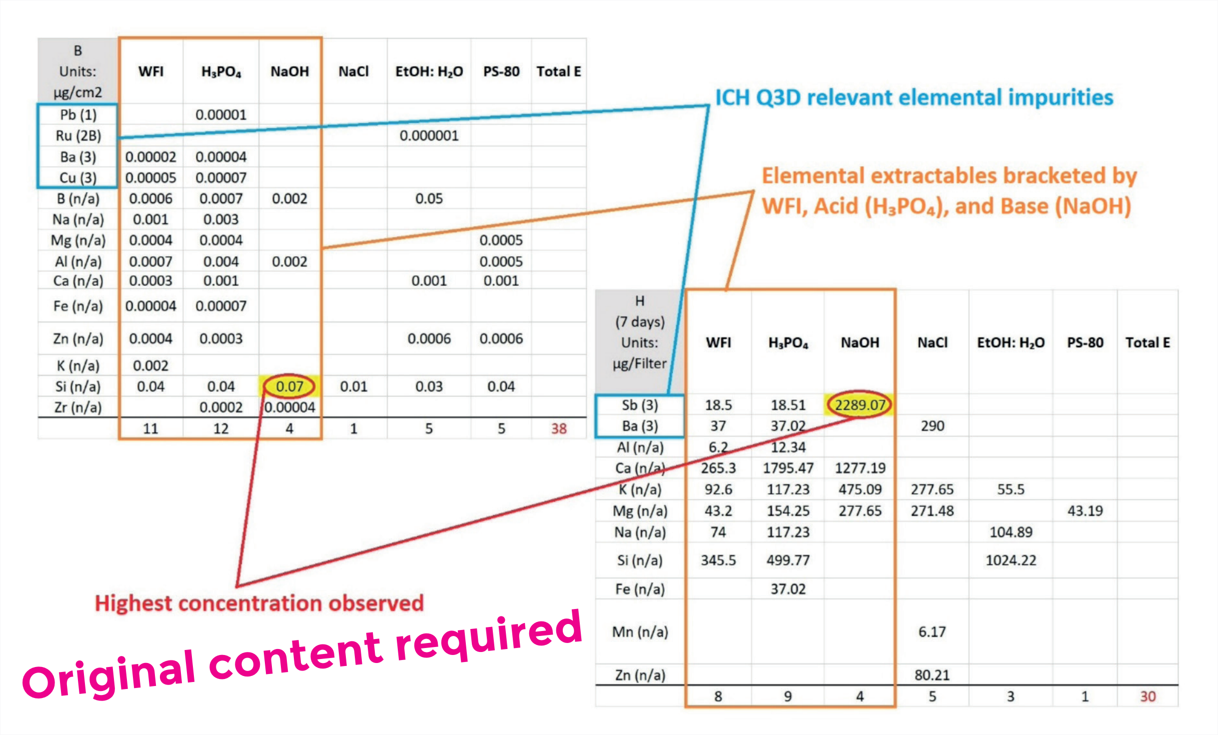
From the original pool of 28 extractables data sets shown in Table 2, only 22 data sets from four suppliers were included in the scope of the elemental extractables data review. This is because one supplier did not provide elemental analysis data and two suppliers provided incomplete data sets where an elemental analysis was not conducted for all six solvents. The following data processing guidelines were utilized for the data review of elemental extractables:

- Only complete data sets with elemental extractables data from all six extraction solvents were included in the scope
- Various component types of bag film, tubing, port, connector and filter were analyzed for the most representative review for elemental extractables
- Time point trends were not analyzed as elemental extractables are reported only for the final extraction time point for each specific component
- If provided by a supplier, additional extractables data from draft USP <665> pH10 base were considered for elemental extractables data analysis only
- For sodium (Na) elemental extractable evaluation purposes only, data from the high ionic strength solvent 5M NaCl were excluded from the evaluation.

The following evaluation criteria were employed. A schematic representation is shown in Figure 4.

- The primary focus was on the evaluation of the ICH Q3D Class 1, 2A, 2B and 3 elements (along with Aluminum (Al), Iron (Fe), Zinc (Zn), Tungsten (W) and Silicon (Si) as these may interact with the product)
- Extraction solvents resulting in the highest level of elemental extractables were identified
- Any 'salting out' effect, chemical incompatibility of a material with an extraction solvent or reported elemental extractables contributed by the extraction solvent (e.g. salt, PS-80 or PS-80 degradants) were taken into consideration to correctly identify and eliminate false positives.

Figure 4: Evaluation criteria used for elemental extractables data analysis



B Units: µg/cm2	WFI	H_3PO_4	NaOH	NaCl	EtOH: H_2O	PS-80	Total E
Pb (1)		0.00001					
Ru (2B)					0.000001		
Ba (3)	0.00002	0.00004					
Cu (3)	0.00005	0.00007					
B (n/a)	0.0006	0.0007	0.002		0.05		
Na (n/a)	0.001	0.003					
Mg (n/a)	0.0004	0.0004				0.0005	
Al (n/a)	0.0007	0.004	0.002			0.0005	
Ca (n/a)	0.0003	0.001			0.001	0.001	
Fe (n/a)	0.00004	0.00007					
Zn (n/a)	0.0004	0.0003			0.0006	0.0006	
K (n/a)	0.002						
Si (n/a)	0.04	0.04	0.07	0.01	0.03	0.04	
Zr (n/a)		0.0002	0.00004				
	11	12	4	1	5	5	38

H (7 days) Units: µg/Filter	WFI	H_3PO_4	NaOH	NaCl	EtOH: H_2O	PS-80	Total E
Sb (3)	18.5	18.51	2289.07				
Ba (3)	37	37.02		290			
Al (n/a)	6.2	12.34					
Ca (n/a)	265.3	1795.47	1277.19				
K (n/a)	92.6	117.23	475.09	277.65	55.5		
Mg (n/a)	43.2	154.25	277.65	271.48		43.19	
Na (n/a)	74	117.23			104.89		
Si (n/a)	345.5	499.77			1024.22		
Fe (n/a)		37.02					
Mn (n/a)				6.17			
Zn (n/a)				80.21			
	8	9	4	5	3	1	30

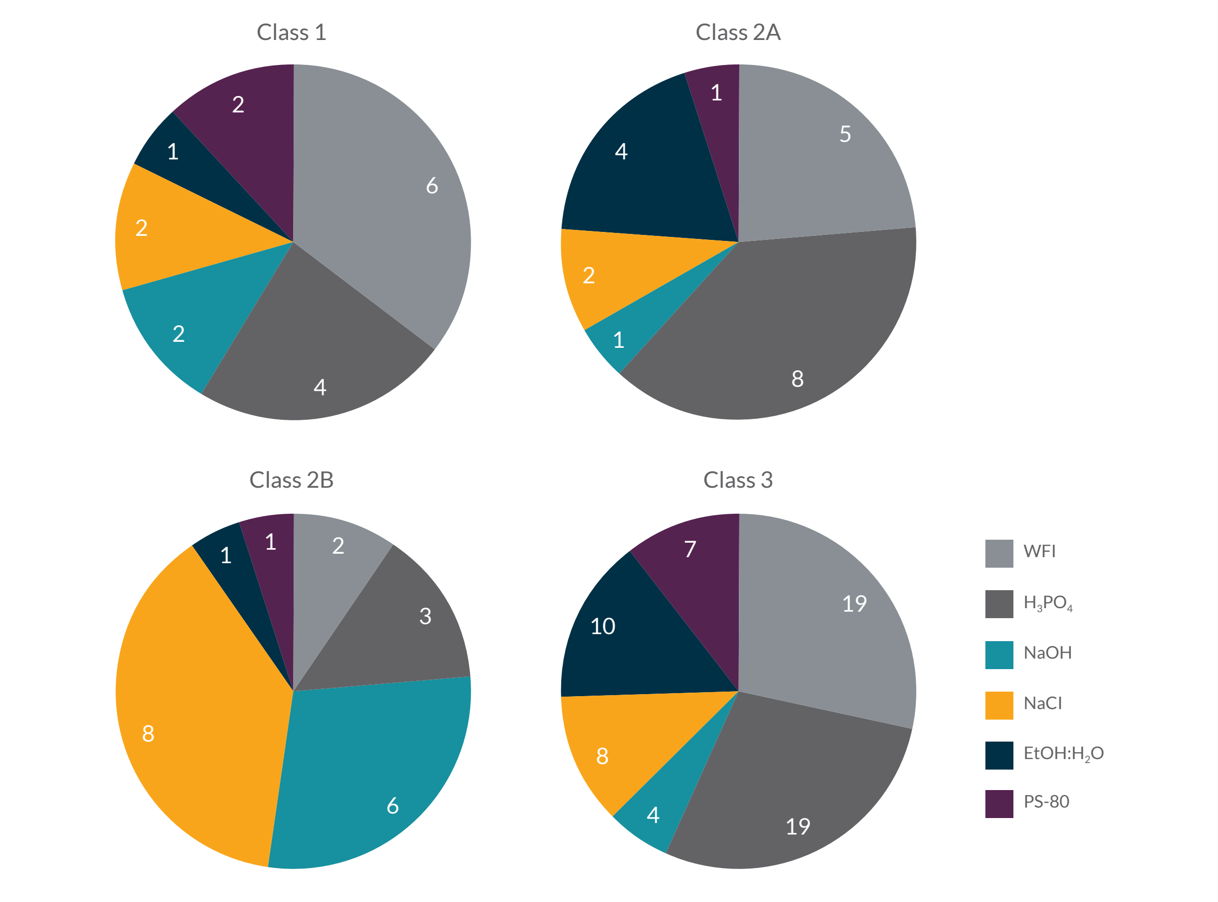
6.2.2 Observations

A total of 529 elemental extractables were reported from 22 data sets, out of which 126 elemental extractables were listed in ICH Q3D as Class 1, 2A, 2B and 3 elemental impurities. The classification of elemental impurities in ICH Q3D guidance[8] is shown in Table 3. The distribution of ICH Q3D-relevant elemental extractables per extraction solvent is shown in Figure 5.

Table 3: ICH Q3D elemental impurity classification

ICH Q3D	Included elemental impurities	Include in risk assessment?
Class 1	As, Pb, Cd, Hg	Yes. These are human toxicants.
Class 2A	Co, Ni, V	Generally considered as route-dependent human toxicants.
Class 2B	Ag, Au, Ir, Os, Pd, Pt, Rh, Ru, Se, Tl	Yes, only if intentionally added.
Class 3	Ba, Cr, Cu, Li, Mo, Sb, Sn	Dependent upon route of administration. Relatively low toxicities.

Figure 5: Distribution of ICH Q3D-relevant elemental extractables in the six extraction solvents



In general, acid and WFI extraction solvents resulted in the highest number of elemental extractables. When a particular elemental extractable exhibited an overlap between acid and WFI solvents, it was observed at a higher concentration level in the acid extraction solvent as compared to WFI. However, in several cases, WFI was observed to provide a higher number of unique elemental extractables as compared to the acid extraction solvent.

Elemental extractables observed in the 50% EtOH:H_2O and PS-80 extraction solvents were generally found to overlap with WFI and acid extraction solvents. With few exceptions (e.g. Si), the concentrations of overlapping elemental impurities observed in WFI and acid extraction solvents were found to be similar or higher than those observed from 50% EtOH:H_2O and PS-80 solvents.

Acid, WFI and base extraction solvents afford greater coverage of the majority of ICH Q3D elemental impurities. In particular, Class 1 (significantly toxic) metals are well captured by acid, WFI and base extraction solvents. Also, the acid extraction solvent appears to be the most suitable for extracting non-ICH Q3D elements such as Al, Fe, Zn, etc.

For overlapping elemental extractable peaks, the highest concentration of elemental extractables appears to be in the base extraction solvent. This was particularly evident for silicon and antimony. However, 5M NaCl and base solvents generally did not extract a wide variety of elemental extractables.

Based on this analysis, the acid, WFI and base extraction solvents (at the final extraction time point) are recommended as the solvents of choice for the elemental extractables study.

6.3 Time point trends

In addition to extraction solvents, all data sets were analyzed for trends across the specified testing time points. No particular increasing or decreasing trend related to any particular solvent or supplier or component was observed. However, it was noted that the first (T0) and the last (T70d) testing time points needed further evaluation to obtain value-added information from these time points.

6.3.1 T0 time point analysis

For the T0 time point analysis, all 28 component data sets were included in the scope of the analysis. Furthermore, when provided by the suppliers, additional data on a second lot of the same component and duplicate data on a single lot of components were considered. These additional datasets were used for time point analysis only, resulting in a total pool of 1,325 data points. The following criteria were established by the review team to identify T0 unique peaks.

Unique peak identification criteria

A peak observed at the first testing time point of T0 would be identified as a valuable unique peak if:

- The given T0 extractable peak is not observed at other time points or is observed at a concentration level that is 50% greater than any other time points across all six extraction solvents
- It is not a PS-80 degradation peak.

Applying Criterion 1

As shown in Table 4, peak 1 was observed in several extracts. The T0 time point from the acid extract yielded a much higher concentration (>50%) than the next highest concentration level detected at any other time point or extract. Therefore, peak 1 is identified as a unique peak per Criterion 1. Similarly, peak 2 from 50% EtOH: H_2O solvent was also identified as a unique peak as it was only detected at T0 time point in the 50% EtOH:H_2O extract.

Table 4: Unique peak identification Criterion 1 for the T0 time point[iii]

Peak	Time point	WFI	5 M NaCl	0.1 M H_3PO_4	0.5 N NaOH	1% PS-80	50% EtOH: H_2O
1	T0	–	0.05	**3.2**	0.07	–	–
	24 hr	1.1	0.99	0.7	1.2	0.38	–
	21 days	0.32	0.85	0.51	0.8	0.41	–
	70 days	0.79	0.55	0.5	0.68	1.1	–
2	T0	–	–	–	–	–	**0.35**
	24 hr	–	–	–	–	–	–
	21 days	–	–	–	–	–	–
	70 days	–	–	–	–	–	–

[iii]example table created for illustrative purposes only

After applying the first criterion to all data sets (a pool of 1,325 peaks), a total of 13 T0 unique peaks were identified.

Applying Criterion 2

For Criterion 2, if the given unique peak was found to originate from the 1% PS-80 extract and the supplier could not confirm elimination of potential PS-80 related degradation peaks, the given peak was eliminated from consideration to avoid false positives in the time point analysis.

After applying Criteria 1 and 2, the pool of unique T0 peaks was reduced to 11 extractable peaks.

At the final step, the potential risk of eliminating T0 time point testing based on the probability of a T0 peak occurrence was determined. The probability of peak occurrence was calculated to be 0.8% (11/1,325×100%) from all included data sets. Based on this analysis, it was determined that the T0 time point added limited value to the standardized extractables testing and eliminating the T0 time point from the testing matrix would present minimal risk of missing material-specific extractables.

6.3.2 70-day time point analysis

For the T70d time point analysis, 13 bag/film data sets (including the second lot and duplicate data sets) were included in the scope for analysis. The following criteria were established by the review team to identify a T70d data point as a unique T70d peak within the data sets, processing a pool of 496 T70d peaks.

Unique peak identification criteria

A peak observed at the last testing time point of (T70d) would be identified as a unique peak if:

- The given T70d peak is not observed at any other timepoint or is observed at a concentration level that is 50% greater than any other time points across all six extraction solvents
- The extractable dataset comprised of all six extraction solvents per the BioPhorum standardized testing protocol. Unlike the T0 time point analysis, the PS-80 extraction datasets were included in the scope for T70d time point analysis. From the given T70d datasets and information provided by the suppliers, it was challenging to distinguish between true extractables and PS-80 degradants. Therefore, all compounds observed at 70 days were considered extractables to represent the worst-case scenario.

As shown in Table 5, peak 1 was observed in several extracts. The T70d time point from the acid extract yielded a much higher concentration (>50%) than the next highest concentration level detected at any other time point or solvent. Therefore, it is identified as a unique peak per Criterion 1. Similarly, peak 2 from the base extract was also considered a unique peak as it was only detected in one extraction solvent at T70d.

Table 5: Unique peak identification criterion 1 for T70d time point[iv]

Peak	Time point	WFI	5 M NaCl	0.1 M H_3PO_4	0.5 N NaOH	1% PS-80	50% EtOH: H_2O
1	T0	–	–	0.03	0.01	–	–
	24 hr	0.02	–	0.05	0.02	–	0.02
	21 days	0.03	–	0.06	0.04	–	0.04
	70 days	0.04	–	**0.15**	0.06	–	0.05
2	T0	–	–	–	–	–	**0.35**
	24 hr	0.05	–	–	0.16	–	–
	21 days	0.07	–	–	0.24	–	–
	70 days	–	–	–	**1.28**	–	–

[iv]example table created for illustrative purposes only

Analysis of the 13 bag/film data sets (including the second lot and duplicate data sets) identified 160 unique T70d peaks, corresponding to 32% of the observed 496 T70d peaks. Further analysis was performed to group the T70d unique peaks into concentration ranges as shown in Table 6. Real-time project examples from BioPhorum member companies were used to demonstrate the significance of each concentration range of unique T70d peaks. The overall consensus among the member companies was that the manufacturing process steps involving the use of large bags (~100L or more) may only be concerned with T70d peaks at concentrations of >1.0μg/cm^2. However, small bags used in the manufacturing process (8 or 16L) may be concerned with peaks at lower concentrations of >0.10μg/cm^2. Based on this analysis, it was concluded that the T70d testing time point provided valuable information and eliminating this point from the standardized testing matrix would present a considerable risk of missing material-specific extractables.

Table 6: Concentration ranges for T70d unique peaks

	All T70d peaks	T70d unique peaks ≥0.01μg/cm^2	T70d unique peaks ≥0.10μg/cm^2	0T70d unique peaks ≥1.00μg/cm^2
Number of peaks	496	160	77	13
% of all T70d peaks	–	32%	16%	3%

7.0

Conclusion

This document presents the results of a systematic review of extractables study data generated per BioPhorum's 2014 standardized extractables testing protocol. The primary aim was to answer the following questions about BioPhorum's standardized test conditions: are all six extraction solvents value-added and are all test time points value-added? The findings outlined in this document form the basis of the updated Biophorum extractables testing protocol.

Extractables data sets were analyzed using a simplified data format to maintain data confidentiality and to ensure a non-biased and scientifically driven review process. Extractable peaks from each solvent and specific test points were assessed to determine if they were unique to that solvent or test point.

The results demonstrated that the high ionic strength extraction solvent (5M NaCl solution) exhibited a minimal number of unique extractables and the surfactant extraction solvent (1% PS-80) demonstrated a high likelihood of providing false-positive data. Moreover, only 0.8% of the total analyzed peaks were found to be unique to the initial testing time point (T0). However, depending on the end-use of the component, the last time point (T70d) was found to provide valuable information about the extractables and their concentration levels.

Based on the comprehensive review of the available data, it is concluded that eliminating the 5M NaCl, 1% PS-80 extraction solvents and the initial testing time point (T0) from the standardized testing matrix presents minimal risk to the determination of a full SUS extractables profile. Also, the distribution of extractable peaks that overlapped with other extraction solvents and/or other extraction time points mitigate the risk of missing any specific extractables.

As for the elemental extractables study solvents, the final extraction time point in 0.1 M H_3PO_4, WFI, and 0.5N NaOH extraction solvents is recommended based on the analysis of high risk associated with the presence of elemental extractables (ICH Q3D relevant elements along with potential product interacting elements) at a high concentration.

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